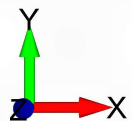
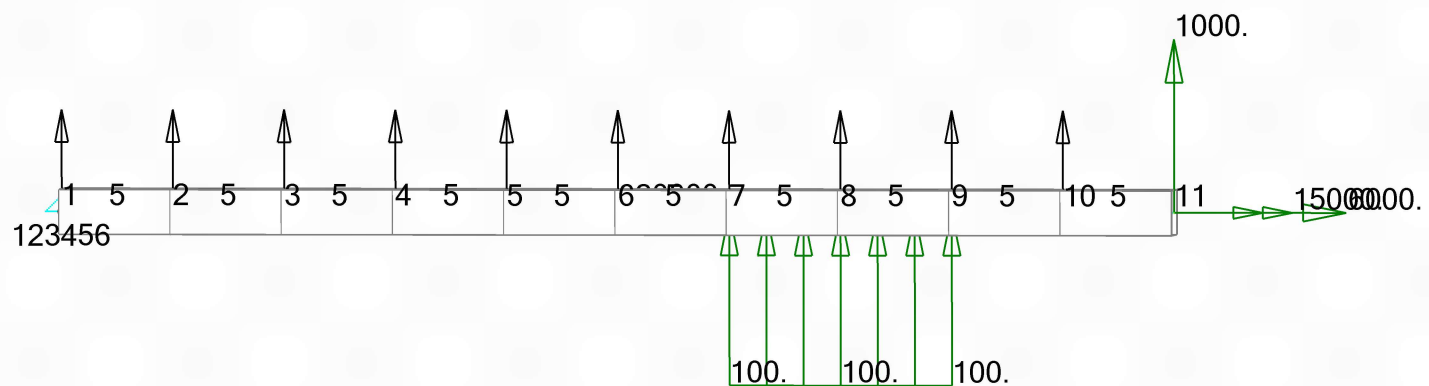


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V: Untitled

L: Static

C: Wall



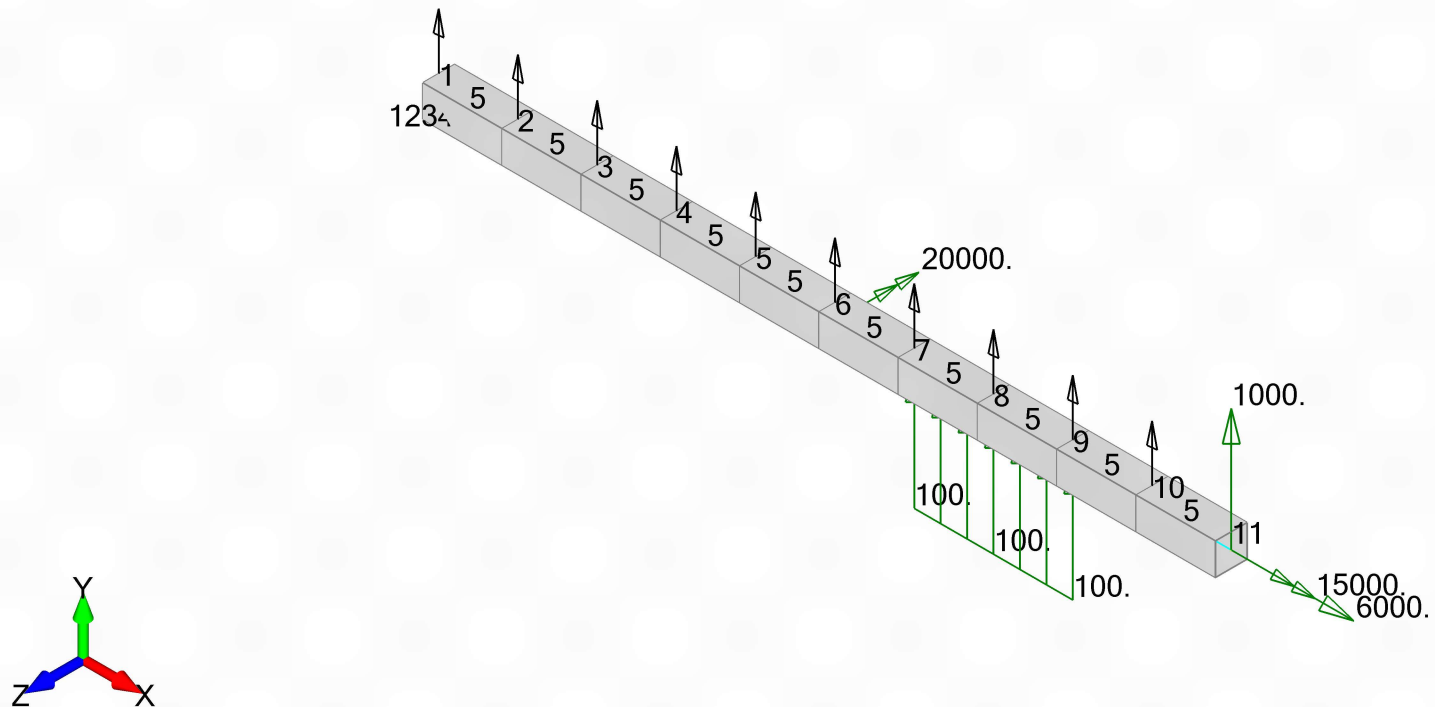
Femap

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L: Static

C: Wall

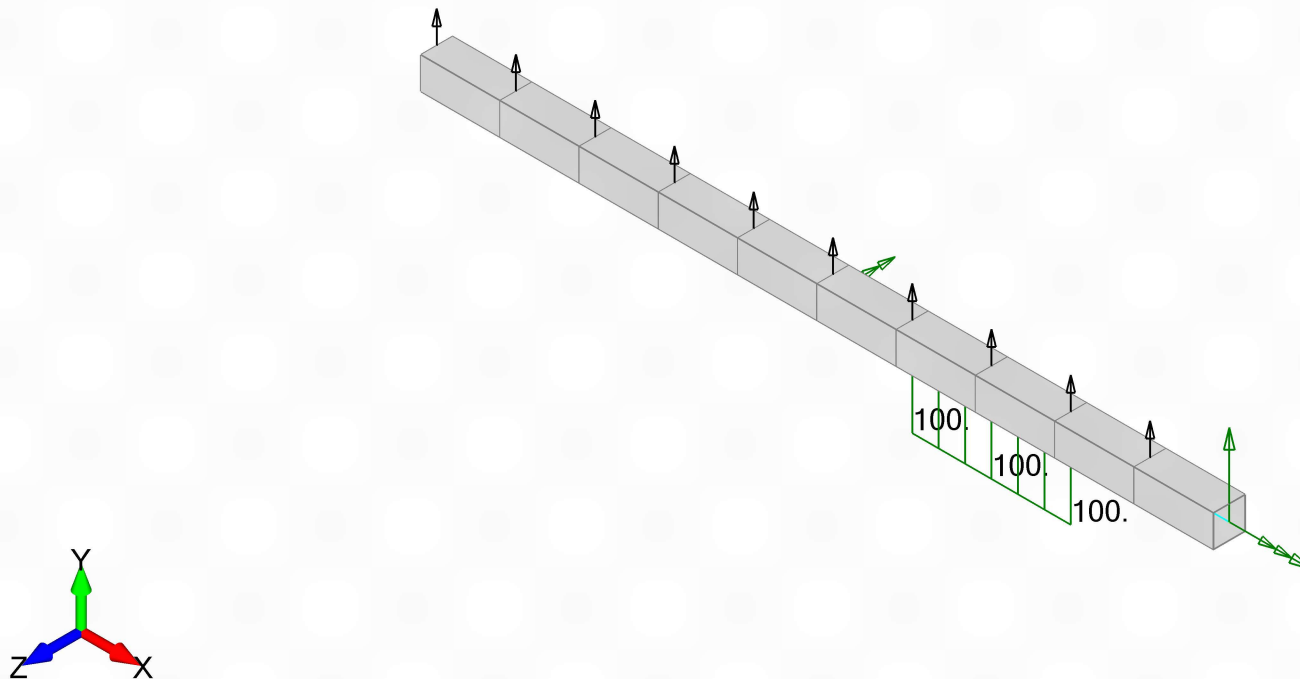


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V: Untitled

L: Static

C: Wall



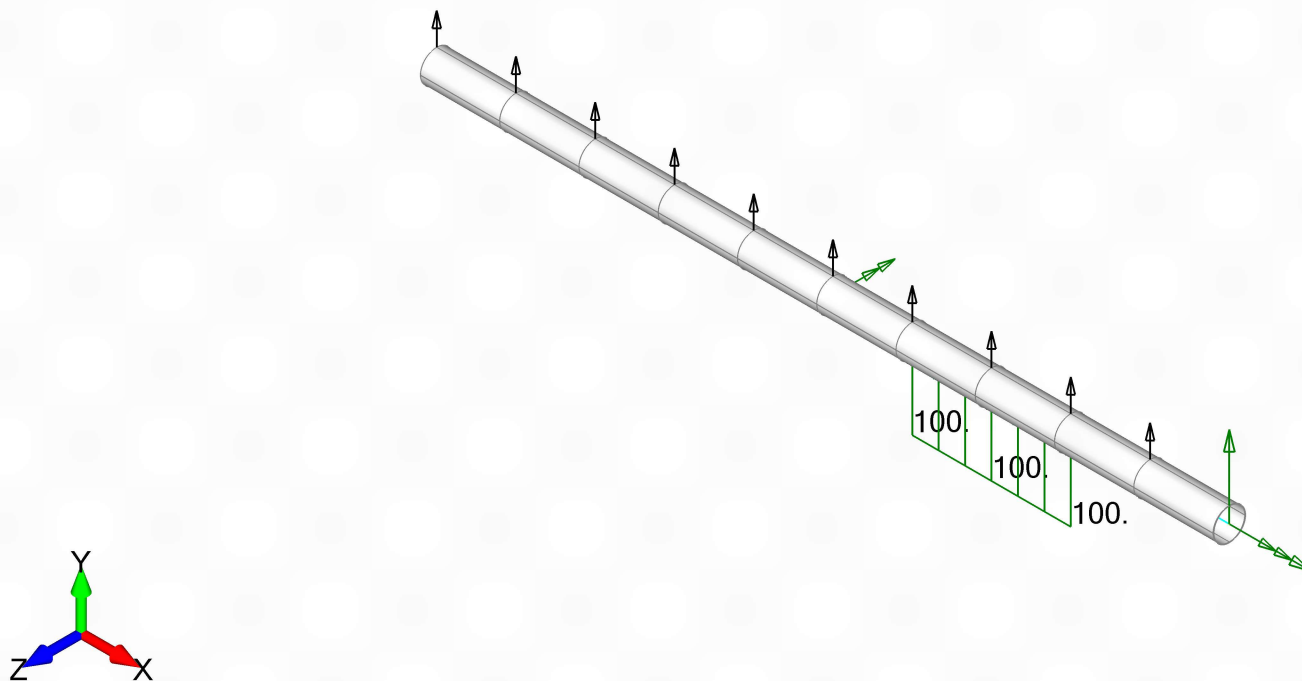
Femap®

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V: Untitled

L: Static

C: Wall



Material 1 - Al			
Type	ISOTROPIC	Color 55	Layer 1
	Density 0.		Damping 0.
STIFFNESS	E 10000000.		G 3846154.
			Nu 0.3
STRENGTH	Tension 0.	Compress 0.	Shear 0.
THERMAL	Alpha 0.	K 0.	SpecHeat 0.
	HtGen 0.		
OPTICAL	Front Off	Reverse Off	

Property 1 : CircularBeam

Material 1 : Al

Circular Bar

=====

AREA

$A = 3.141593$

MOMENTS OF INERTIA

$I_{zz} = 0.785398$

$I_{yy} = 0.785398$

$I_{zy} = 0.$

PRINCIPAL INERTIAS

Max, $I_1 = 0.785398$

Min, $I_2 = 0.785398$

Polar, $I_p = 1.570796$

Angle, $\alpha = 0.$

RADIUS OF GYRATION

$y = 0.5$

$z = 0.5$

CENTROID FROM ORIGIN

$H = 1.$

$V = 1.$

STRESS RECOVERY LOCATIONS

1 $H = 1.$

$V = 0.$

2 $H = 2.$

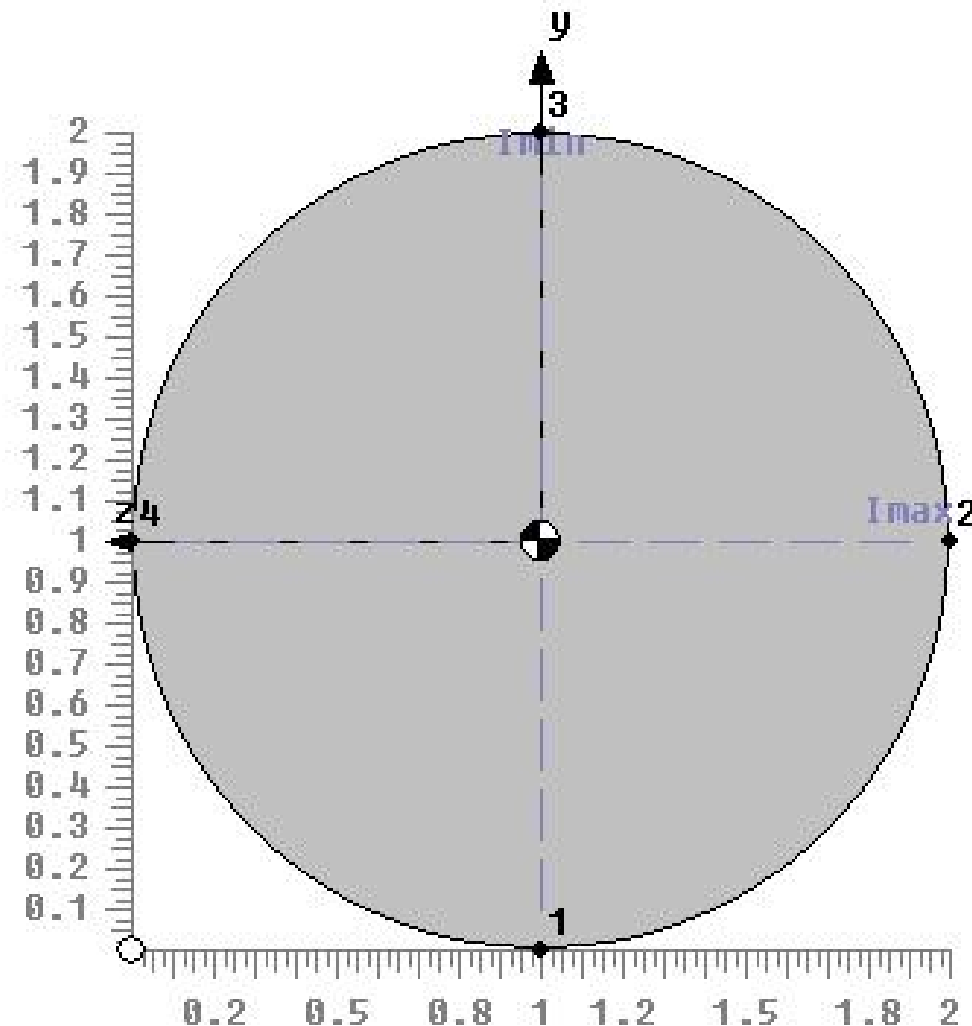
$V = 1.$

3 $H = 1.$

$V = 2.$

4 $H = 0.$

$V = 1.$



Node 1

Output Set

1 : Simcenter NASTRAN Case 1

Value =	0.		Output Vector	1 : Total Translation
Value =	0.		Output Vector	2 : T1 Translation
Value =	0.		Output Vector	3 : T2 Translation
Value =	0.		Output Vector	4 : T3 Translation
Value =	0.		Output Vector	5 : Total Rotation
Value =	0.		Output Vector	6 : R1 Rotation
Value =	0.		Output Vector	7 : R2 Rotation
Value =	0.		Output Vector	8 : R3 Rotation
Value =	0.		Output Vector	41 : Total Applied Force
Value =	0.		Output Vector	42 : T1 Applied Force
Value =	0.		Output Vector	43 : T2 Applied Force
Value =	0.		Output Vector	44 : T3 Applied Force
Value =	0.		Output Vector	45 : Total Applied Moment
Value =	0.		Output Vector	46 : R1 Applied Moment
Value =	0.		Output Vector	47 : R2 Applied Moment
Value =	0.		Output Vector	48 : R3 Applied Moment
Value =	6324.555		Output Vector	51 : Total Constraint Force
Value =	-6000.		Output Vector	52 : T1 Constraint Force
Value =	-2000.		Output Vector	53 : T2 Constraint Force
Value =	0.		Output Vector	54 : T3 Constraint Force
Value =	66708.32		Output Vector	55 : Total Constraint Moment
Value =	-15000.		Output Vector	56 : R1 Constraint Moment
Value =	0.		Output Vector	57 : R2 Constraint Moment
Value =	-65000.		Output Vector	58 : R3 Constraint Moment

Node 2

Output Set 1 : Simcenter NASTRAN Case 1

Value =	0.0990841		Output Vector	1 : Total Translation
Value =	0.00095493		Output Vector	2 : T1 Translation
Value =	0.0990795		Output Vector	3 : T2 Translation
Value =	0.		Output Vector	4 : T3 Translation
Value =	0.0401671		Output Vector	5 : Total Rotation
Value =	0.0124245		Output Vector	6 : R1 Rotation
Value =	0.		Output Vector	7 : R2 Rotation
Value =	0.0381972		Output Vector	8 : R3 Rotation
Value =	0.		Output Vector	41 : Total Applied Force
Value =	0.		Output Vector	42 : T1 Applied Force
Value =	0.		Output Vector	43 : T2 Applied Force
Value =	0.		Output Vector	44 : T3 Applied Force
Value =	0.		Output Vector	45 : Total Applied Moment
Value =	0.		Output Vector	46 : R1 Applied Moment
Value =	0.		Output Vector	47 : R2 Applied Moment
Value =	0.		Output Vector	48 : R3 Applied Moment
Value =	0.		Output Vector	51 : Total Constraint Force
Value =	0.		Output Vector	52 : T1 Constraint Force
Value =	0.		Output Vector	53 : T2 Constraint Force
Value =	0.		Output Vector	54 : T3 Constraint Force
Value =	0.		Output Vector	55 : Total Constraint Moment
Value =	0.		Output Vector	56 : R1 Constraint Moment
Value =	0.		Output Vector	57 : R2 Constraint Moment
Value =	0.		Output Vector	58 : R3 Constraint Moment

Node 9

Output Set

1 : Simcenter NASTRAN Case 1

Value =	4.203398		Output Vector
Value =	0.00763944		Output Vector
Value =	4.203391		Output Vector
Value =	0.		Output Vector
Value =	0.194895		Output Vector
Value =	0.0993964		Output Vector
Value =	0.		Output Vector
Value =	0.167643		Output Vector
Value =	250.		Output Vector
Value =	0.		Output Vector
Value =	250.		Output Vector
Value =	0.		Output Vector
Value =	208.3333		Output Vector
Value =	0.		Output Vector
Value =	0.		Output Vector
Value =	-208.3333		Output Vector
Value =	0.		Output Vector
Value =	0.		Output Vector
Value =	0.		Output Vector
Value =	0.		Output Vector
Value =	0.		Output Vector
Value =	0.		Output Vector
Value =	0.		Output Vector

1 : Total Translation
2 : T1 Translation
3 : T2 Translation
4 : T3 Translation
5 : Total Rotation
6 : R1 Rotation
7 : R2 Rotation
8 : R3 Rotation
41 : Total Applied Force
42 : T1 Applied Force
43 : T2 Applied Force
44 : T3 Applied Force
45 : Total Applied Moment
46 : R1 Applied Moment
47 : R2 Applied Moment
48 : R3 Applied Moment
51 : Total Constraint Force
52 : T1 Constraint Force
53 : T2 Constraint Force
54 : T3 Constraint Force
55 : Total Constraint Moment
56 : R1 Constraint Moment
57 : R2 Constraint Moment
58 : R3 Constraint Moment

Node 6

Output Set

1 : Simcenter NASTRAN Case 1

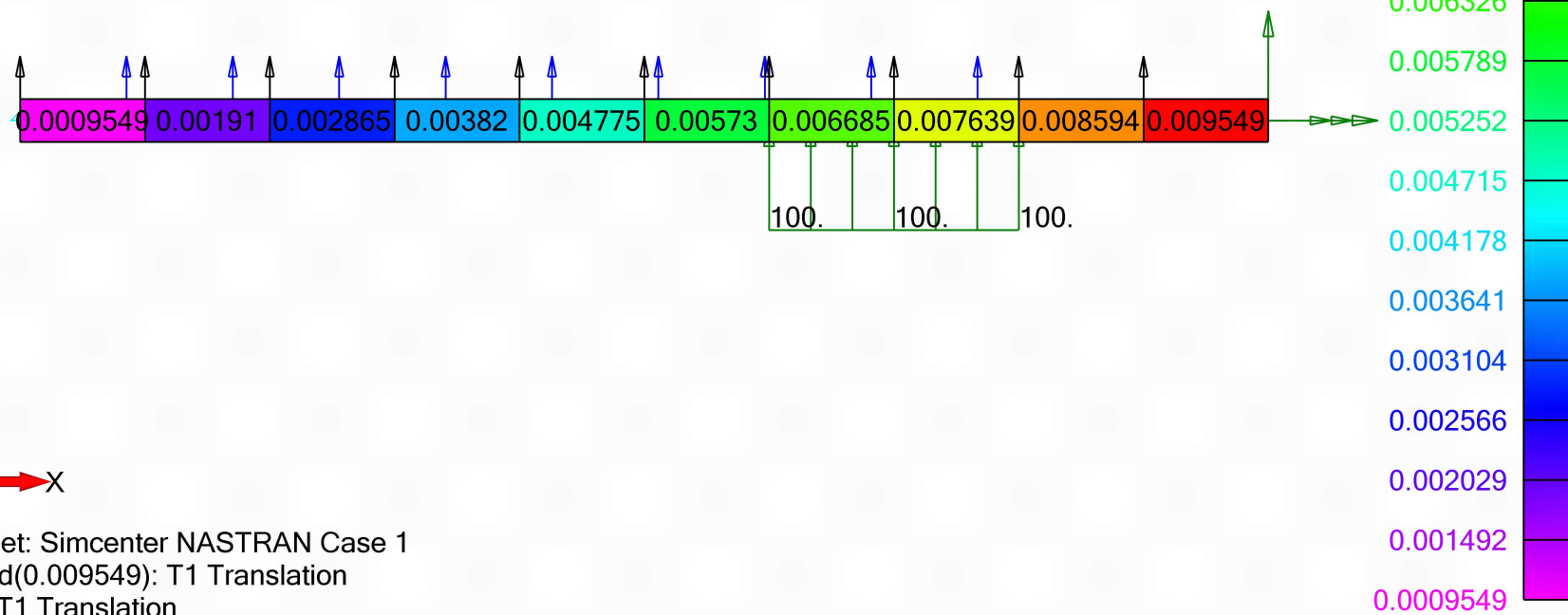
Value =	1.927798		Output Vector	1 : Total Translation
Value =	0.00477465		Output Vector	2 : T1 Translation
Value =	1.927792		Output Vector	3 : T2 Translation
Value =	0.		Output Vector	4 : T3 Translation
Value =	0.141671		Output Vector	5 : Total Rotation
Value =	0.0621227		Output Vector	6 : R1 Rotation
Value =	0.		Output Vector	7 : R2 Rotation
Value =	0.127324		Output Vector	8 : R3 Rotation
Value =	0.		Output Vector	41 : Total Applied Force
Value =	0.		Output Vector	42 : T1 Applied Force
Value =	0.		Output Vector	43 : T2 Applied Force
Value =	0.		Output Vector	44 : T3 Applied Force
Value =	20000.		Output Vector	45 : Total Applied Moment
Value =	0.		Output Vector	46 : R1 Applied Moment
Value =	0.		Output Vector	47 : R2 Applied Moment
Value =	-20000.		Output Vector	48 : R3 Applied Moment
Value =	0.		Output Vector	51 : Total Constraint Force
Value =	0.		Output Vector	52 : T1 Constraint Force
Value =	0.		Output Vector	53 : T2 Constraint Force
Value =	0.		Output Vector	54 : T3 Constraint Force
Value =	0.		Output Vector	55 : Total Constraint Moment
Value =	0.		Output Vector	56 : R1 Constraint Moment
Value =	0.		Output Vector	57 : R2 Constraint Moment
Value =	0.		Output Vector	58 : R3 Constraint Moment

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V: Untitled

L: Static

C: Wall

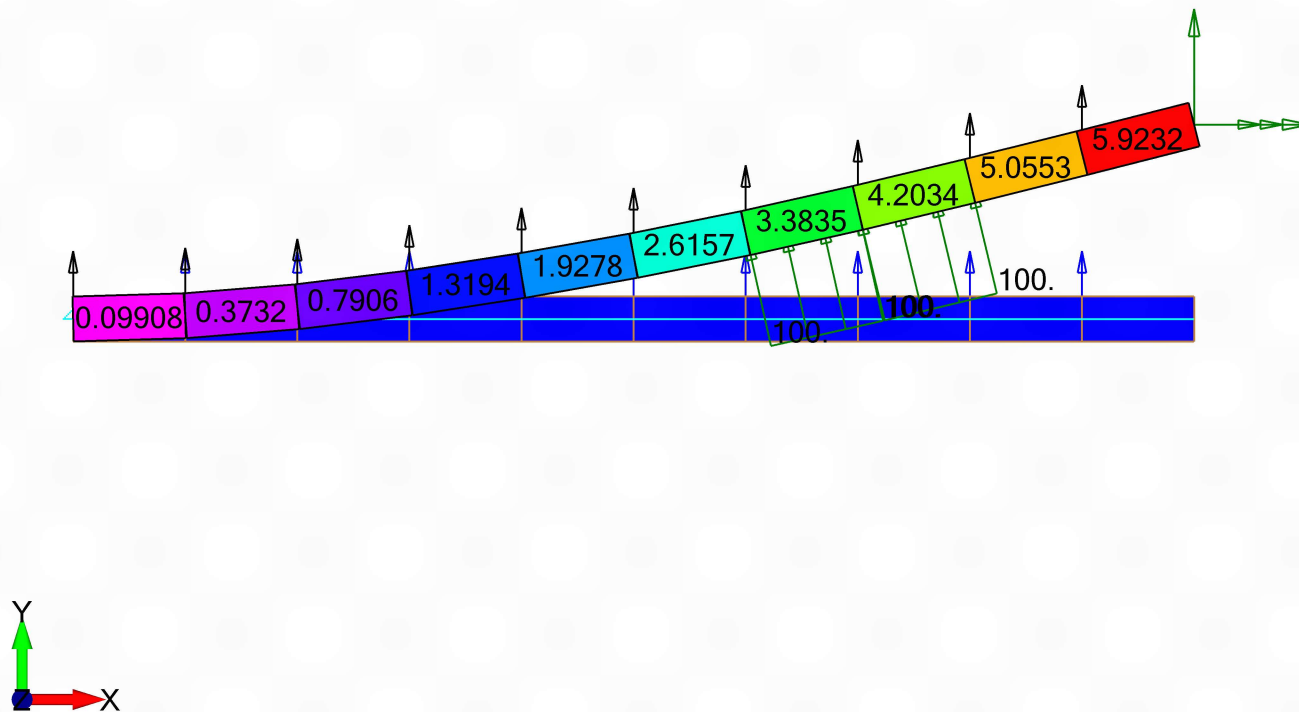


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V: Untitled

L: Static

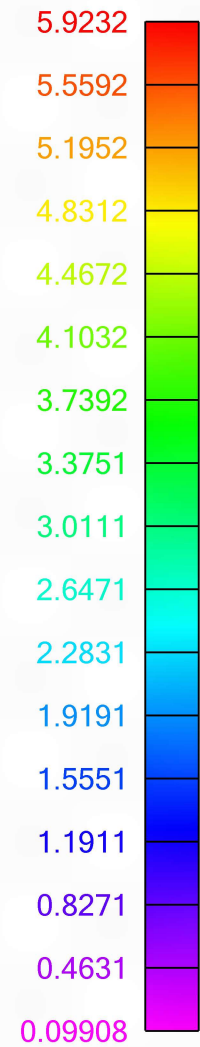
C: Wall



Output Set: Simcenter NASTRAN Case 1

Deformed(5.9232): T2 Translation

Criteria: T2 Translation

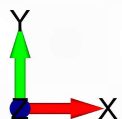
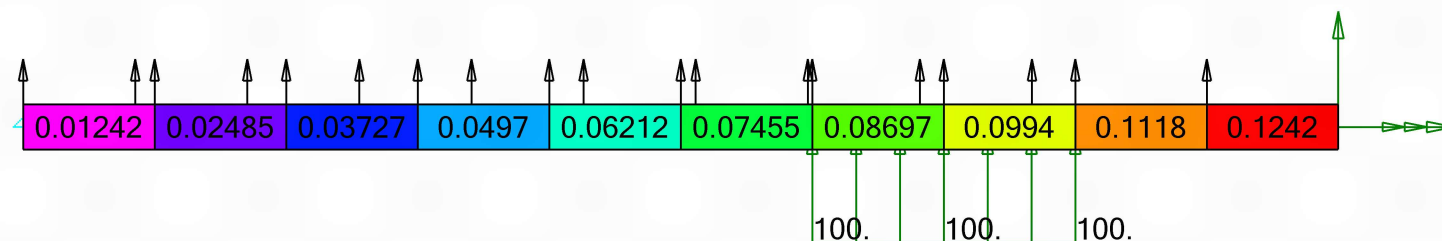


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V: Untitled

L: Static

C: Wall



Output Set: Simcenter NASTRAN Case 1

Deformed(0.1242): R1 Rotation

Criteria: R1 Rotation

0.1242

0.1173

0.1103

0.1033

0.09629

0.0893

0.08231

0.07532

0.06833

0.06135

0.05436

0.04737

0.04038

0.03339

0.0264

0.01941

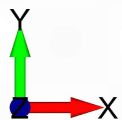
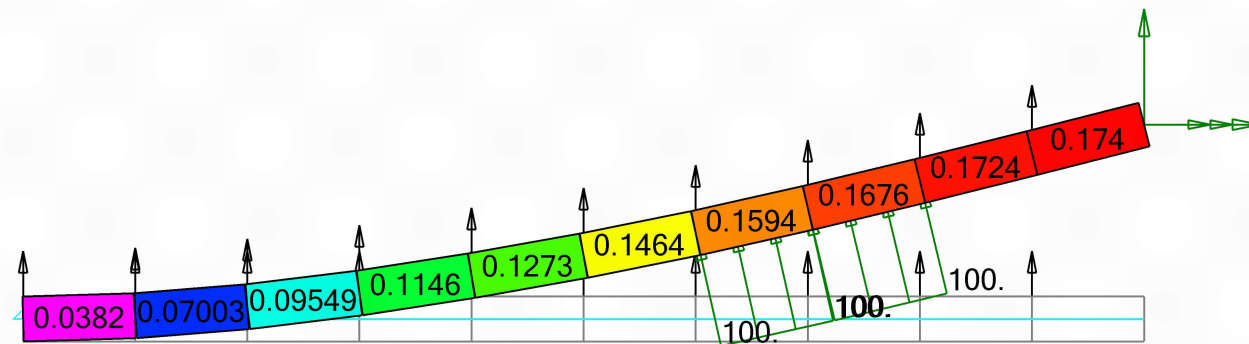
0.01242

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L: Static

C: Wall



Output Set: Simcenter NASTRAN Case 1

Deformed(5.9232): T2 Translation

Criteria: R3 Rotation

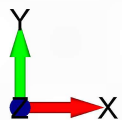
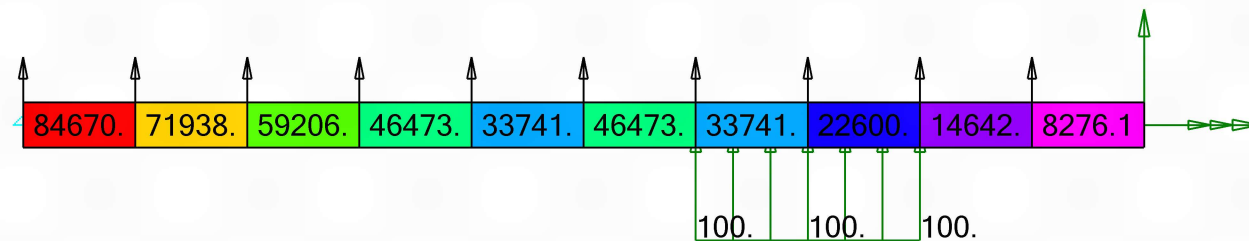


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L: Static

C: Wall



Output Set: Simcenter NASTRAN Case 1
Criteria: Beam EndA Max Comb Stress

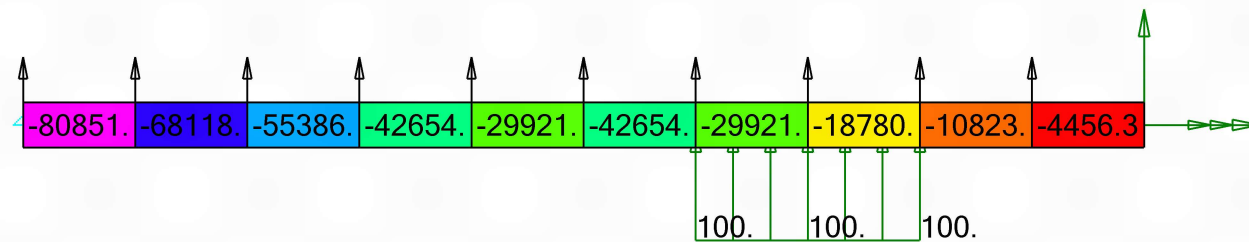


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L: Static

C: Wall



-4456.3

-9231.0

-14006.0

-18780.0

-23555.0

-28330.0

-33104.0

-37879.0

-42654.0

-47428.0

-52203.0

-56977.0

-61752.0

-66527.0

-71301.0

-76076.0

-80851.3

Output Set: Simcenter NASTRAN Case 1

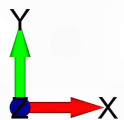
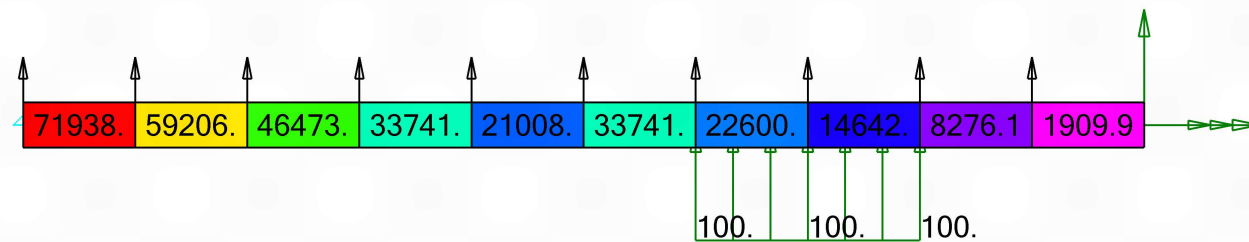
Criteria: Beam EndA Min Comb Stress

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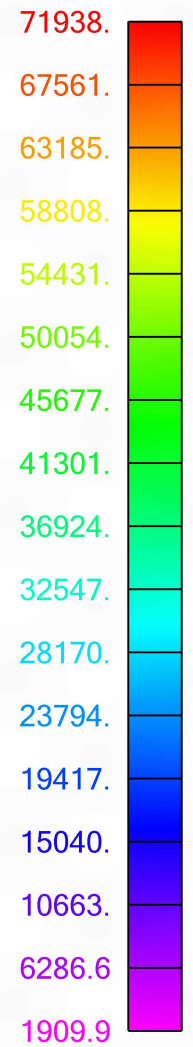
V: Untitled

L: Static

C: Wall



Output Set: Simcenter NASTRAN Case 1
Criteria: Beam EndB Max Comb Stress

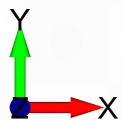
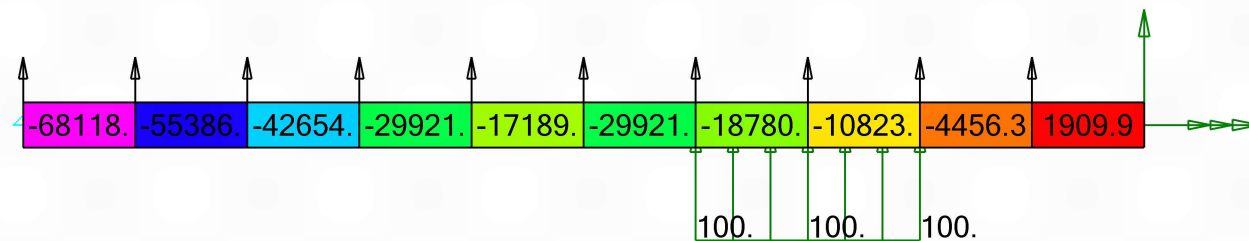


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V: Untitled

L: Static

C: Wall



Output Set: Simcenter NASTRAN Case 1

Criteria: Beam EndB Min Comb Stress

1909.9

-2466.9

-6843.7

-11220.

-15597.

-19974.

-24351.

-28727.

-33104.

-37481.

-41858.

-46235.

-50611.

-54988.

-59365.

-63742.

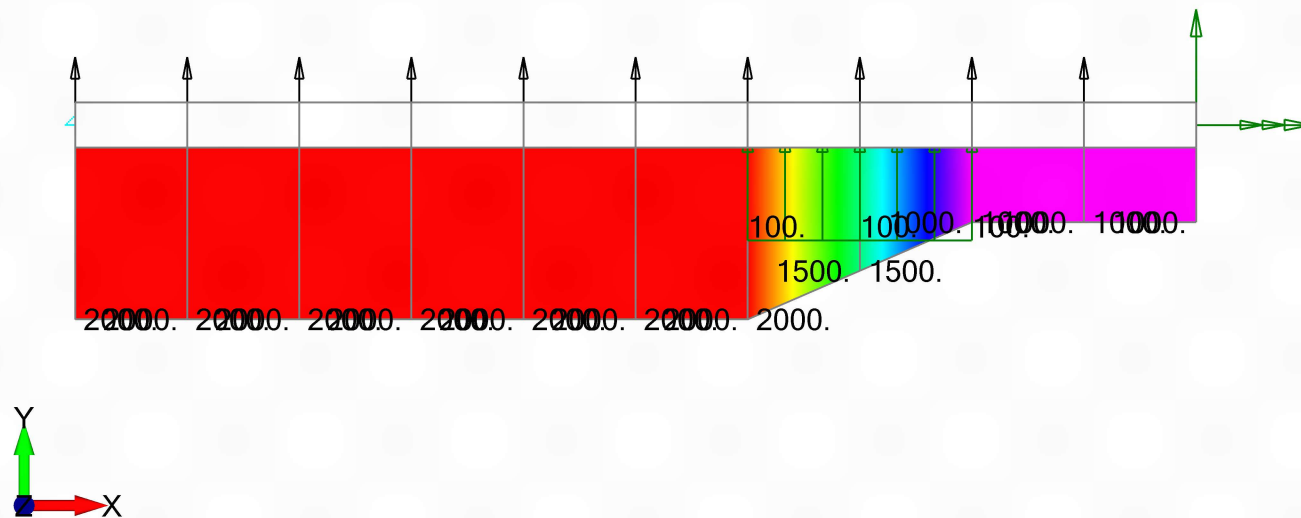
-68118.

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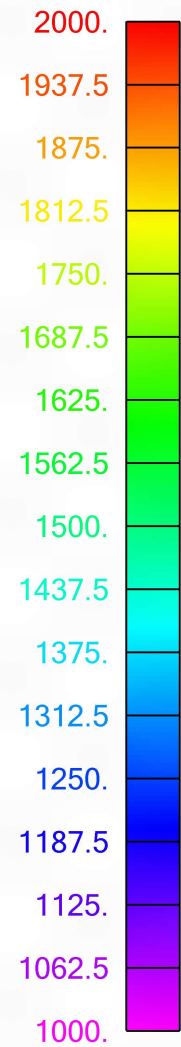
V: Untitled

L: Static

C: Wall



Output Set: Simcenter NASTRAN Case 1
Contour: Beam EndA PI1 Shear Force

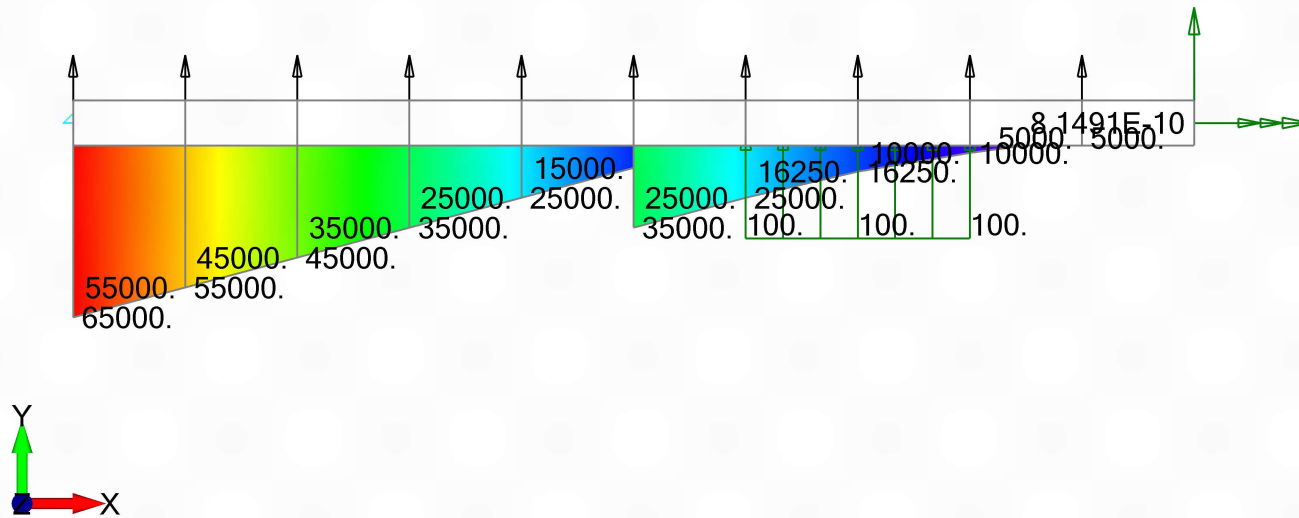


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L: Static

C: Wall



Output Set: Simcenter NASTRAN Case 1
Contour: Beam EndA Plane1 Moment